

Database Systems Lab

Project Update — Milestones 2, 3, 4 & 5

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All groups are required to complete the following milestones in order. Read through the full document carefully before starting work.

Milestone 2 — ERD Design & Normalization

You have already completed your ER Diagram and initial schema. The next step is to apply normalization formally and document it as part of your version-controlled repository.

Step 1 — Apply Normalization

Review every table in your schema and apply the normalization process in sequence: First Normal Form (1NF), then Second Normal Form (2NF), then Third Normal Form (3NF). For each normal form, write down what the issue was, what change you made, and why. If your schema already satisfies a normal form without any changes, you are still required to write a short justification confirming this and explaining why no change was needed. Stating that a table is already in 3NF without any supporting explanation will not be accepted.

Step 2 — Remove Duplicates

Check all tables for redundant columns, repeated data, or overlapping attributes. Remove or restructure as necessary and note what was changed.

Step 3 — Update Your ERD

Reflect all normalization changes in your ER Diagram. Ensure all primary keys, foreign keys, relationships, and cardinalities are accurately shown in the final version.

Step 4 — Commit to GitHub

Add a normalization document to your repository, either as NORMALIZATION.md or as a dedicated section within your README. Commit the updated ERD diagram alongside it. Your commit message must clearly describe what was updated, for example:

- M2: Applied 2NF and 3NF normalization, updated ERD and schema

Milestone 3 — Dataset Preprocessing

Step 1 — Prepare Your Dataset

If your project uses a real dataset (CSV, API, or otherwise), clean it. Remove duplicates, fix formatting inconsistencies, handle null or missing values, and ensure data types are consistent throughout.

If your project uses dummy data, you are required to generate structured synthetic data. It must be realistic and meaningful enough to properly populate your database. Tools you may use include Python with the Faker library, Mockaroo, or Excel. In either case, the data must be

sufficient to populate all core tables with a minimum of 50 to 100 rows per table.

Step 2 — Define Your Dataflow

Every group must include a Dataflow section in their documentation. This should clearly explain where data enters your system, how it moves through the database (which tables receive data, which tables depend on others), and what comes out — whether that is query results, reports, exports, or AI model inputs, depending on your project. This description must be written specifically for your project. A generic or copy-pasted description will not be accepted.

Step 3 — Export Clean CSV Files

Once your data is ready, export it as CSV files, one file per table. These files will be used in Milestone 5 to load data into your database.

Step 4 — Commit to GitHub

Push all cleaned or synthetic CSV files and your dataflow description to the repository.

- M3: Synthetic data generated; dataflow documented

Milestone 4 — Database Setup (DDL)

Write all CREATE TABLE statements based on your finalized normalized schema. Every table must include a primary key, appropriate foreign key references, NOT NULL constraints where required, and UNIQUE or CHECK constraints where applicable. Add indexes on foreign key columns and any other columns you expect to query frequently. Once done, verify your schema in MySQL Workbench and confirm the EER diagram matches your DDL scripts.

Commit all DDL scripts to GitHub with the following message:

- M4: DDL scripts added, EER diagram verified

Milestone 5 — Data Population (DML)

Load your data into the database using either LOAD DATA INFILE with your prepared CSV files, or INSERT statements — both approaches are acceptable. Demonstrate at least one UPDATE and one DELETE operation with a WHERE condition.

After loading your data, run the following validation queries and include the output (screenshots or a commented .sql file):

- COUNT (*) for each table to confirm row counts
- A NULL check on key columns
- A JOIN-based check to confirm foreign key integrity is intact

Commit all DML scripts and validation output to GitHub:

- M5: Data populated validation queries added

Submission Requirements

Every group must complete both of the following by the deadline.

1. GitHub Repository

Push all updates to your repository. It must contain:

- Updated ERD diagram
- Normalization document with justifications (1NF to 3NF)
- Dataflow description
- Cleaned or synthetic CSV files
- DDL scripts (CREATE TABLE statements)
- DML scripts with validation query outputs
- A clear commit history with at least one commit per milestone

2. PDF Upload (submit here in Classroom)

Compile and upload a single PDF version control doc containing:

- Group name and both member names
- Updated ERD diagram
- Normalization walkthrough (1NF to 3NF with justifications)
- Dataflow description
- GitHub repository link

PDF filename format: GroupName_Version_Control_DBLab.pdf

Please Note

- Both group members must have contributed commits to the repository. Submissions showing commits from only one member will be flagged.
- Normalization without written justification will not receive full marks, even if the schema is technically correct.
- Synthetic data must be structured and realistic. Placeholder or random data with no logical relation to your schema is not acceptable.
- The dataflow description must reflect your specific project. Generic write-ups will not be accepted.